Student Information

Name: Ahmed Yasser Abu Elkher Salah

Faculty Code: 2420232 Mark: ..... / 10

Assignment 2

For your chosen system or project, it is essential to provide a comprehensive analysis that encompasses all critical aspects of the system you intend to design and develop. *Your submission should include the following details*:

- ✓ **System Name**: Clearly define the name of your system to establish its identity.

> **Answer:** **Clinic Management System (SmartCare CMS)** - *Project #7 (Tutorial Sheet 6).*
> An integrated desktop healthcare application built using Java Swing GUI (NetBeans IDE) & MySQL Database.

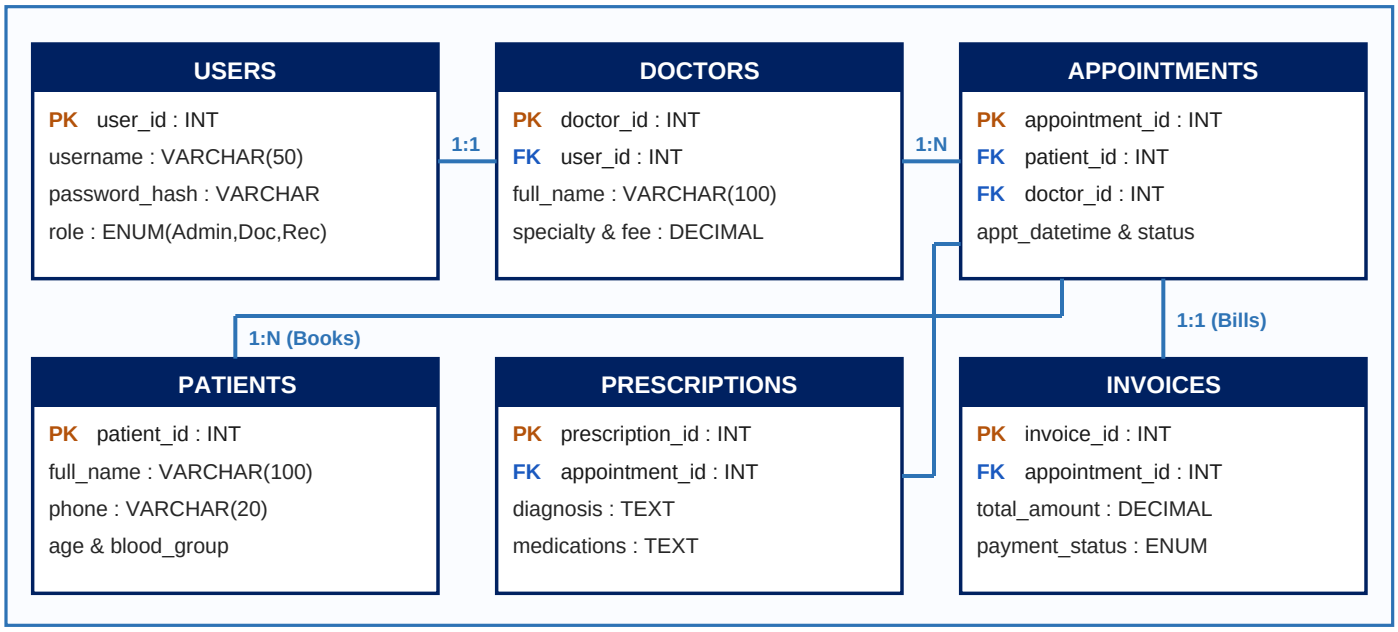
- ✓ **System Aim & Services**: Outline the primary objectives of the system and the range of services it will offer to its users.

> **Primary Aim:** Automate clinic workflows by replacing paper files with a fast, accurate Java desktop system.
>
> **1. Patient Management:** Register, update, and search patient profiles, medical history, and blood group.
>
> **2. Doctor & Appointment Scheduling:** Manage doctor shifts, specialties, and book visits without conflicts.
>
> **3. EMR & Prescription Service:** Record clinical diagnoses, vital signs, and issue digital prescriptions.
>
> **4. Billing & Invoicing Service:** Calculate consultation fees, generate receipts, and track daily revenue.

- ✓ **Brief Overview of the System**: Provide a succinct description that captures the essence of the system, including its functionality and intended user experience.

> Built following the Java Swing Containment Hierarchy (Lecture 5), the GUI uses a top-level JFrame container with modular JPanel screens, JTable data grids, JTextField/JComboBox inputs, JButton actions, and JOptionPane confirmation dialogs. The Receptionist registers patients and books appointments; the Doctor views the daily queue and writes prescriptions; and the system generates the invoice. Developed using the Agile SDLC methodology across 6 phases: Requirements, Design, GUI/JDBC Coding, Testing, Deployment, Maintenance.

✓ **Database Diagram**: Include a detailed diagram illustrating the database structure, showcasing how data is organized and interconnected within the system.



✓ **Additional Elements**: Consider incorporating other pertinent information that may enhance the understanding of the system, such as user roles, security measures, and system architecture.

1. User Roles (RBAC): Admin (manages staff & reports), Doctor (diagnoses & prescriptions), Receptionist (bookings & billing).

2. Security Measures: SHA-256 password hashing, JDBC PreparedStatement (prevents SQL Injection), JOptionPane validation.

3. System Architecture: 3-Tier MVC Desktop Architecture (Java Swing GUI View -> Controllers/Models -> JDBC MySQL DAO).

ClinicDatabaseConnection.java — NetBeans IDE (Java Swing + MySQL JDBC)

```java
import java.sql.Connection;
import java.sql.DriverManager;
import javax.swing.JOptionPane;

public class ClinicDatabaseConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/smartcare_clinic_db";
    public static Connection getConnection() {
        try {
            return DriverManager.getConnection(URL, "root", "");
        } catch (Exception ex) {
            JOptionPane.showMessageDialog(null, "DB Error: " + ex.getMessage());
            return null;
        }
    }
}
```

By thoroughly addressing these elements, you will create a well-rounded analysis that clearly communicates the vision and framework of your project.